

Final Project Milestone 1 Literature Review

1. <https://arxiv.org/abs/2301.11125>

A benchmark for toxic comment classification on Civil Comments dataset (Duchêne et al., 2023)

This paper presents a comprehensive benchmark for toxic comment detection using the Civil Comments dataset, a large multi-label dataset derived from Jigsaw's unintended bias in toxicity classification challenge. The authors compare a wide range of neural architectures, including transformer-based models (BERT, RoBERTa, DistilBERT, ALBERT, HateBERT, XLNet), RNNs (BiLSTM and BiGRU), and vision-inspired architectures (ResNet, Compact Convolutional Transformer). The models were trained and evaluated using consistent preprocessing, training protocols, and metrics (AUROC, F1, bias AUC measures), ensuring fair comparison. The dataset was preprocessed by removing HTML tags, URLs, diacritics, and extra whitespace, and by converting all text to lowercase. Due to severe class imbalance, the authors down-sampled non-toxic examples and applied several loss functions such as Binary Cross Entropy, Focal Loss, and Positive Weighted Binary Cross Entropy to improve model learning stability.

Results show that transformer-based models consistently outperform CNN and CCT architectures, while RNNs, particularly BiLSTM, offer a strong trade-off between performance and inference speed. Among transformers, RoBERTa with Focal Loss achieved the highest AUROC and least bias, whereas DistilBERT balanced strong performance with low inference time. Despite their effectiveness, all models exhibited identity-associated bias, especially toward groups such as “Black,” “Muslim,” and “Homosexual.” The study concludes that while BERT-family models remain state-of-the-art for toxic comment classification, careful attention to bias mitigation and inference efficiency is crucial for real-world moderation systems.

2. <https://arxiv.org/abs/2012.10289>

HateXplain: A Benchmark Dataset for Explainable Hate Speech Detection (Mathew et al. , 2021)

Mathew et al. (2021) introduce *HateXplain*, the first large-scale dataset designed not only to classify online posts as hateful, offensive, or normal, but also to capture *why* human annotators made those judgments. About 20 000 tweets and Gab posts were labeled through Amazon Mechanical Turk, with each entry including the target community and text spans (“rationales”) that justify the label. The authors trained several neural architectures—CNN-GRU, BiRNN, BiRNN-Attention, and BERT—both with and without these human-provided rationales. Their results show that models using rationales achieved slightly higher accuracy and substantially less unintended bias toward identity groups (for example, African, Muslim, or Jewish communities). However, high accuracy did not always mean good explainability: models could predict the correct label but rely on token patterns different from those

humans found important. Overall, HateXplain demonstrated that adding human rationales improves fairness and interpretability, offering a valuable benchmark for future hate-speech research.

3. <https://arxiv.org/html/2501.01256v1>

Digital Guardians: Can GPT-4, Perspective API, and Moderation API reliably detect hate speech in reader comments of German online newspapers?

This paper evaluates the effectiveness of pretrained large language models (LLMs) and moderation APIs in detecting hate speech in German for specific context: reader comments of online newspapers. The study directly compares three main approaches: OpenAI's GPT-4o model, Google's Perspective API, and the OpenAI Moderation API. These solutions were tested against a specialized, fine-tuned BERT-based model (the HOCON34k classifier) that serves as the baseline, using the HOCON34k dataset of annotated German comments. For the GPT-4o model, the researchers specifically tested its performance using Zero-Shot, One-Shot, and Few-Shot prompting techniques to see how it performed with varying levels of context. For the OpenAI Moderation API, the overall "True/False" score was used. And for the Perspective API, its probability output was mapped to a binary class using three different thresholds.

The study found that the GPT-4o model was highly effective, even outperforming the specialized baseline model. Interestingly, providing more examples did not necessarily improve GPT-4o's performance; Zero-Shot learning (with no examples) yielded the best results for metrics like accuracy and precision, while One-Shot learning (with a single example) achieved the highest recall and F2-score. The One-Shot approach, in particular, surpassed the fine-tuned BERT baseline by approximately 5 percentage points on the study's combined performance metric (the Champion-Challenger score). This suggests that general-purpose, pre-trained LLMs like GPT-4o can be more effective at hate speech detection than specialized, fine-tuned models, even with minimal prompting.